

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EC-5004-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Communication Networks and Transmission Lines

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions out of eight questions.

ii) All questions carry equal marks.

iii) Assume suitable data, if required.

1. a) Discuss the iterative impedance and characteristic impedance of symmetrical two port network.
b) Explain the design of symmetrical attenuators. Also discuss their working principle.
2. a) How many types of the network we can use in transmission lines? Explain each one of them.
b) Give some of the differences between active and passive filters with examples.
3. a) Explain and derive an equation for the single stub matching and double stub matching?
b) Write an introductory note on m-derived filters and Chebyshev approximation.
4. a) Derive a common relation for phase velocity, group velocity and free space velocity.
b) Explain the expressions for the input impedance of open and short circuited dissipation less line.

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[2]

5.
 - a) Write an operating principle of Bott duffin method. Explain the concept of minimum positive real function.
 - b) Discuss the Foster and Causer Network. Explain the Brune's method and synthesis coefficient.
6.
 - a) The characteristics impedance of a 805m long transmission line is $4\angle -23.2^\circ \Omega$ the attenuation constant is 74.5×10^{-6} Np/m and the phase shift constant is 174×10^{-6} rad/m at 5kHz. Calculate the line parameters R, L, G and C per meter and the phase velocity on the line.
 - b) State the significance of crystal filters in communication system. Write a brief note on frequency and phase distortion.
7.
 - a) Explain the reflection coefficient of different transmission lines. Explain briefly about open circuit and short circuit line.
 - b) Write some points on the losses in transmission lines. Explain the T and Z equivalents of a line.
8. Write short notes (any four):
 - a) Stub Matching
 - b) RLC Network synthesis
 - c) Micro strip lines
 - d) Lattice Networks
 - e) Composite Filters

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